Basic Probability

Exam code: 0607

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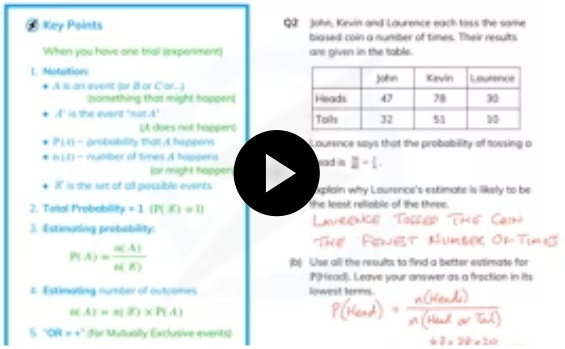

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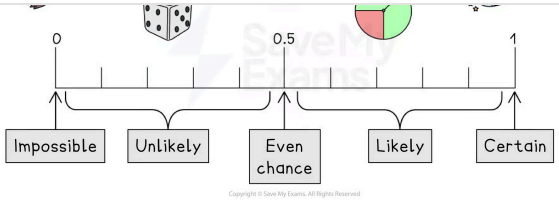
Basic probability

What is probability?

- Probability describes the **likelihood** of something happening
  - In real-life you might use words such as **impossible**, **unlikely** and **certain**
- In maths we use the **probability scale** to describe probability
  - This means giving it a number between **0 and 1**
    - **0** means **impossible**
    - Between **0 and 0.5** means **unlikely**
    - **0.5** means **even chance**
    - Between **0.5 and 1** means **likely**
    - **1** means **certain**
- Probabilities can be given as **fractions**, **decimals** or **percentages**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand



Probability scale

What key words and terminology are used in probability?

- An **experiment** is an activity that is **repeated** to produce a set of **results**
  - Results can be observed (seen) or recorded
  - Each repeat is called a **trial**

- An **outcome** is a possible result of a trial
- An **event** is an outcome (or a collection of outcomes)
  - For example:
    - a dice lands on a six
    - a dice lands on an even number

  - Events are usually given capital letters
  - n(*A*) is the number of possible outcomes from event *A*
    - *A* = a dice lands on an even number (2, 4 or 6)
    - n(*A*) = 3

- A **sample space** is the set of **all** possible outcomes of an experiment
  - It can be represented as a **list** or a **table**

- The **probability** of event *A* is written P(*A*)
- An event is said to be **fair** if there is an **equal chance** of achieving each outcome
  - If there is not an equal chance, the event is **biased**
  - For example, a fair coin has an equal chance of landing on heads or tails

How do I calculate basic probabilities?

- If all outcomes are **equally likely** then the probability for each outcome is the same
  - The probability for each outcome is $\frac{1}{\text{Total number of outcomes}}$
    - If there are 50 marbles in a bag then the probability of selecting a specific one is $\frac{1}{50}$

- The **theoretical probability** of an event can be calculated by dividing the number of outcomes of that event by the total number of outcomes
  - $\mathrm{P}(A) = \frac{\text{Total number of outcomes for the event}}{\text{Total number of outcomes}}$
  - This can be calculated without actually doing the experiment

How do I find missing probabilities?

- The probabilities of **all** the outcomes **add up to 1**
  - If you have a table of probabilities with one **missing**, find it by making them all add up to 1
- The **complement of event *A*** is the event where ***A* does not happen**
  - This can be thought of as **not *A***
  - P(event does **not** happen) = 1 - P(event **does** happen)
    - For example, if the probability of rain is 0.3, then the probability of **not** rain is 1 - 0.3 = 0.7

What are mutually exclusive events?

- Two events are **mutually exclusive** if they **cannot both happen at once**
  - When rolling a dice, the events "getting a prime number" and "getting a 6" are mutually exclusive (as 6 is not prime)
- If *A* and *B* are mutually exclusive events, then the probability of **either A or B happening** is **P(A) + P(B)**
- **Complementary events** are mutually exclusive

Examiner Tips and Tricks

If you are not told in the question how to leave your answer, then fractions are best for probabilities.

Worked Example

Emilia is using a spinner that has outcomes and probabilities as shown in the table.

Outcome	Blue	Yellow	Green	Red	Purple
Probability		0.2	0.1		0.4

The spinner has an equal chance of landing on blue or red.

(a) Complete the probability table.

Answer:

The probabilities of **all** the outcomes **should add** up to **1**

1 - 0.2 - 0.1 - 0.4 = 0.3

The probability that it lands on blue or red is 0.3
As the probabilities of blue and red are **equal** you can halve this to get each probability

Outcome	Blue	Yellow	Green	Red	Purple
Probability	**0.15**	0.2	0.1	**0.15**	0.4

(b) Find the probability that the spinner lands on green or purple.

Answer:

As the spinner cannot land on green and purple at the same time they are **mutually exclusive**
This means you can **add** their probabilities together

$$0.1 + 0.4 = 0.5$$

P(Green or Purple) = 0.5

(c) Find the probability that the spinner does not land on yellow.

Answer:

The probability of **not** landing on yellow is equal to 1 minus the probability of landing on yellow

$$1 - 0.2 = 0.8$$

P(Not Yellow) = 0.8

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